13.7.1 Three stages of Antigen – Antibody Reactions

a) Primary stage

The reactions between antigen and antibody occur in three stages. The primary stage is the initial interaction between the two without any visible effects. This reaction is rapid and obeys the general laws of physical chemistry and thermodynamics. The reaction is reversible. The combination between antigen and antibody is effected by the weaker intermolecular forces such as electrostatic forces, hydrogen bonds, Van der Waals forces and hydrophobic forces.

b) Secondary stage

The primary stage is followed by the secondary stage leading to demonstrable events such as precipitation, agglutination, lysis of cells, killing of live antigens, neutralization of motile organisms, complement fixation and enhancement of phagocytosis.

c) Tertiary stage

Some antigen-antibody reactions occurring in vivo initiate chain reactions that lead to neutralization or destruction of injurious antigens or to tissue damage. These are the tertiary reactions and include humoral immunity against infectious diseases as well as clinical allergy and other immunological diseases.

13.7.2 General Features of Antigen – Antibody Reactions

Antigen-antibody reactions have the following general characteristics:

1. The antigen-antibody reaction is specific. An antigen combines only with its homologous antibody and vice versa.
2. An entire molecule reacts and not fragments.
3. There is no denaturation of the antigen or the antibody during the reaction.
4. The combination occurs at the surface.
5. TBoth antigen and antibody participate in the formation of agglutinates or precipitates.

13.7.3 Measurement of Antigen and Antibody

Many methods are available for the measurement of antigens and antibodies participating in the primary, secondary and tertiary reactions. Measurement may be in terms of mass (Example: mg Nitrogen) or more commonly as units or titre. The antibody titre of a serum is the highest dilution of the serum which gives an observable reaction with the antigen in the particular test. The titre of a serum is influenced by the nature and quantity of the antigen and the type and conditions of the test. Antigens may also be titrated against sera.

The various tests used for detection of antigen and antibodies are given below:

1. Precipitation tests
2. Agglutintion tests
3. Complement Fixation test
4. Immunofluorescence
5. Radio immuno assay
6. Enzyme linked immuno sorbent assay

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